$Area = \frac{1}{2} \times Base \times Height$ sq. units

That is, $A = \frac{1}{2} \times b \times h$ sq. units

where, b is base and h is height of the triangle.

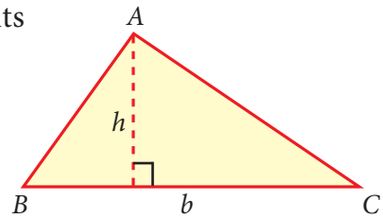


Fig. 7.1

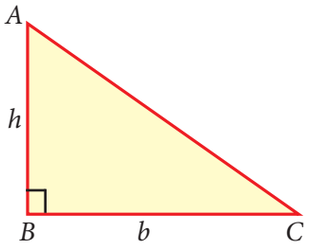


Fig. 7.2

From the above, we know how to find the area of a triangle when its 'base' and 'height' (that is altitude) are given.

7.2 Heron's Formula

How will you find the area of a triangle, if the height is not known but the lengths of the three sides are known?

For this, Heron has given a formula to find the area of a triangle.

If a, b and c are the sides of a triangle, then the area of a triangle $= \sqrt{s(s-a)(s-b)(s-c)}$ sq.units.
where $s = \frac{a+b+c}{2}$, 's' is the semi-perimeter (that is half of the perimeter) of the triangle.

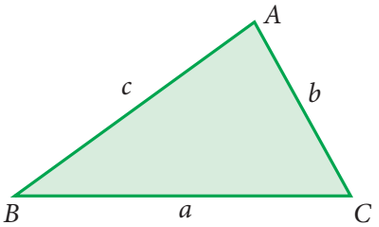


Fig. 7.3

Note

If we assume that the sides are of equal length that is $a = b = c$, then Heron's formula will be $\frac{\sqrt{3}}{4}a^2$ sq.units, which is the area of an equilateral triangle.

Example 7.1 The lengths of sides of a triangular field are 28 m, 15 m and 41 m. Calculate the area of the field. Find the cost of levelling the field at the rate of ₹ 20 per m^2.

Solution

Let $a = 28\,m$, $b = 15\,m$ and $c = 41\,m$

Then, $s = \frac{a+b+c}{2} = \frac{28+15+41}{2} = \frac{84}{2} = 42\ m$

Area of triangular field $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{42(42-28)(42-15)(42-41)}$

$= \sqrt{42 \times 14 \times 27 \times 1}$

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